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Review

Global Geographical and Temporal Patterns of Seasonal Influenza and Associated Climatic Factors

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Understanding geographical and temporal patterns of seasonal influenza can help strengthen influenza surveillance to early detect epidemics and inform influenza prevention and control programs. We examined variations in spatiotemporal patterns of seasonal influenza in different global regions and explored climatic factors that influence differences in influenza seasonality, through a systematic review of peer-reviewed publications. The literature search was conducted to identify original studies published between January 2005 and November 2016. Studies were selected using predetermined inclusion and exclusion criteria. The primary outcome was influenza cases; additional outcomes included seasonal or temporal patterns of influenza seasonality, study regions (temperate or tropical), and associated climatic factors. Of the 2,160 records identified in the selection process, 36 eligible studies were included. There were significant differences in influenza seasonality in terms of the time of onset, duration, number of peaks, and amplitude of epidemics between temperate and tropical/subtropical regions. Different viral types, cocirculation of influenza viruses, and climatic factors, especially temperature and absolute humidity, contributed to the variations in spatiotemporal patterns of seasonal influenza. The findings reported in this review could inform global surveillance of seasonal influenza and influenza prevention and control measures such as vaccination recommendations for different regions.

climatic factors; geographic and temporal patterns; influenza viral strains; seasonal influenza

Abbreviations: AH, absolute humidity; E-SE, east and southeast; H1N1, influenza A H1N1; H3N2, influenza A H3N2; WHO, World Health Organization.

INTRODUCTION

Influenza is a highly contagious viral infection that usually presents as an annual seasonal epidemic or occasional pandemic (1). Currently, there are two major types of the influenza viruses, influenza A and influenza B, causing human infections, with influenza A accountable for more deaths than influenza B (1, 2). The global annual incidence of influenza is substantially higher in children (20%–30%) compared with adults (5%–10%) (3). The epidemics of seasonal influenza are estimated to cause 3 million to 5 million cases of severe illness and to lead to 290,000–650,000 deaths worldwide annually (4). These figures highlight the burden of influenza and its implications for high social and economic costs (3, 4). The World Health Organization (WHO) encourages global efforts on influenza prevention and vaccination, especially for people at a higher risk of serious complications and death (3).

Despite being a vaccine-preventable disease (5), its rapid transmission and relatively high incidence during seasonal epidemics and pandemics make influenza one of the major health concerns worldwide. Furthermore, the virulent nature of some influenza viruses, the occasional zoonotic outbreaks, and the viruses' ability to undergo important genetic changes through antigenic shifts make it challenging to predict the next influenza epidemic and to develop effective prevention measures (6, 7). The global influenza activity reports released by WHO suggest variations in the number of annual peaks, the onset of epidemics, and the duration of seasonal influenza transmission in the northern and southern hemispheres and various geographical locations in each hemisphere (8, 9).

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Although there are many subtypes of influenza A virus, the major subtypes responsible for annual seasonal influenza epidemics in humans include influenza A H1N1 (H1N1) and influenza A H3N2 (H3N2). The patterns of influenza transmission also vary according to the dominant viral subtype (9).

Understanding geographical and temporal variations of seasonal influenza can help strengthen the existing influenza surveillance to detect early onset of epidemics, plan control measures to reduce the incidence in vulnerable populations, and provide recommendations on relevant prevention strategies (e.g., vaccination plans) in the general population. Despite numerous empirical studies (including some population-based observational studies) in which researchers addressed the patterns of influenza seasonality in various countries or regions of the world and covered different time periods, including after the 2009–2010 influenza pandemic, existing literature remains inconclusive in describing long-term patterns of seasonal influenza across different global regions, especially the transmission patterns in the tropics and subtropics (10–13). Latitudinal variations in seasonal activity of influenza and the mechanisms explaining the different patterns of influenza seasonality across temperate and tropical regions at global level were examined in 2 review articles. The authors suggested that influenza seasonality in tropical localities and the link between different viral epidemics and climatic and environmental factors need to be addressed and well understood (10, 14). To date, to our knowledge, no research has been developed to systematically analyze the variations in the patterns of seasonal influenza around the globe in consideration of the dynamics of virus strains and possible climatic contributors. In this study, we examined the common geographical and temporal (long-term) patterns of seasonal influenza and explored variations in the patterns of seasonal influenza along with their inter-relationships with climatic factors through a systematic review of internationally published research and health reports on the relevant topics.

METHODS

Search strategy and selection criteria

In November 2016, we searched the PubMed database for published literature relevant to the study topic. Possible mechanisms driving global influenza seasonality were explored in a 2011 review article that covered a wide range of publications from the 1960s up to 2010 (14); thus, we limited the present study to articles published in January 2005 through November 2016 to update information relevant to the topic and compare the findings with existing evidence. This time span also allowed us to assess studies relating to the 2009 influenza pandemic. The searches were conducted using different combinations of key words, including "influenza seasonality OR seasonal influenza" with various combinations of search terms like "geographical differences"; "geographical variations"; "geographical patterns"; "spatial variations"; "temporal differences"; "temporal variations"; "temporal patterns"; "changes over time"; "long term trends"; "climatic factors," and "climatic variations". The same search strategies and key words were repeated in different databases, including Google Scholar, Science Direct, and Scopus, in December 2016. The reference lists of the articles included in this review were manually searched for additional publications on the relevant topics.

Inclusion criteria

Original observational studies focusing on human population data and published in English within the predetermined period were included for this systematic review. In the initial article search and screening process, if the title or abstract of an article contained the terms "seasonal influenza" and either "geographic" or "temporal" patterns (or any of the interchangeable key words listed in the previous paragraph), it was included for further full-text reading. Influenza activity or patterns had to be included as 1 of the key outcomes. Many studies for which data were obtained from national or international surveillance systems (e.g., WHO FluNet) in which cases of influenza-like illness and/or acute respiratory infections were routinely reported and monitored. In such studies, confirmation of influenza cases should be established to allow a breakdown of analysis on influenza patterns.

The inclusion criteria were not limited to any particular region of the world or country, nor to any particular population age group. Data and findings at the individual patient and population levels were sought in relation to the key words of interest. Longitudinal studies in which data for more than 1 influenza season were used and with clearly defined seasonality (i.e., onset, duration, and peaks of epidemics) were considered. Geographic variations were defined as any differences in seasonal patterns of influenza epidemics between northern and southern hemispheres, different continents, diverse global regions (temperate, subtropical, and tropical), various countries, or smaller studies including only 1 country or state across multiple latitudinal gradients.

Exclusion criteria

Studies focusing on nonhuman zoonotic influenza epidemics or viral antigenic characteristics or pathological processes of influenza were excluded from this review. Interventional studies and clinical trials examining the effects of drugs and other interventions, such as vaccinations, on influenza epidemics were excluded because these did not meet of the objectives for the review. Studies limited to a single influenza season for 1 local area or city were also excluded from the review unless the topics were related to specific pandemics or large-scale outbreaks. Studies investigating the 2009–2010 pandemic were excluded because the pandemic did not follow the usual seasonal patterns of influenza. The relevance to the review topic (i.e., spatial and/or temporal patterns of seasonal influenza) was also taken into consideration. Thus, publications that did not provide clear definitions of either temporal or geographic patterns of influenza seasonality were eliminated.

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Study selection

Both study authors were involved in determining the key words and the literature search and data extraction process. The initial literature search was conducted by the first author. Each key-word combination used in the search process was shared and verified with the second author and repeated in different database searches. All records from different data sources were retrieved and uploaded to an EndNote (Clarivate Analytics, Philadelphia, PA) library. In this process, duplicated records obtained from different sources were removed.

The inclusion and exclusion criteria for eligibility screening were discussed between the authors and determined at the title and abstract-reading stages. The decision to exclude each article was made according to the criteria. The next step in the selection process was undertaken by reading the full-text manuscripts to address all inclusion and exclusion criteria and the study aims. The full-text articles selected to be reviewed were critically assessed, using critical appraisal tools (the Preferred Reporting Items for Systematic Reviews and Meta-Analyses checklist and the quality assessment tool for quantitative studies developed by the McMaster University) (15, 16), for the risk of bias and the quality of information.

The data of 4 studies were collected from influenza-like illness surveillance. However, specimens of influenza were tested in 3 studies to allow separate analyses of influenza patterns (17–19). Only 1 study used the national influenza-like illness surveillance in Vietnam data for analysis (20). Considering the unique environmental and climatic diversity within the country and the similar patterns between influenza-like illness and influenza epidemics over time observed in other Asian countries, this study was retained for analysis. The assessment results were verified between the authors. Any disagreements between the researchers arising during study selection and critical appraisal processes were carefully discussed to resolve any inconsistencies in making the final decision.

Data analysis

The main outcome used for this review was seasonal influenza. However, studies extended to explore large-scale influenza outbreaks or pandemics were also included. The majority of the studies were considered methodologically sound, having used appropriate study designs and rigorous data collection and analysis methods (rated "strong" according to both critical appraisal tools). Overall, 2 studies were rated "weak" against most of the assessment criteria and were excluded from the review. However, they were used to affirm the key findings reported in the included studies. Data extracted for this review included author, year of study, study setting, population and sample size, study design, source of data, case ascertainment, statistical analysis methods, main spatial or temporal patterns of influenza, and associated climatic factors. Data were extracted from the included publications and displayed on an electronic spreadsheet (Excel; Microsoft, Redmond, WA). The outcome measure was based on confirmed cases of influenza or influenza-like illness reported by the authors of each study.

RESULTS

A total of 2,614 studies initially were retrieved from the databases. After removing the duplicates from different data sources, the remaining 2,160 articles were screened through title, abstract, and further full-text reading along with the application of inclusion and exclusion criteria at each step of elimination. Figure 1 illustrates the details in the process of study selection. The decisions at each step of the selection process were discussed and validated between the authors. At the end of the entire process, 36 publications were included for this systematic review.

A summary of the key extracted components of the included studies is presented in Table 1. The study publication years spanned from 2007 to 2016. The most common range of study period was from the year 2000 onward ($n = 26$ articles, 72.2%). More than 40% ($n = 17$) of the included studies involved large-scale data from multiple countries or even across continents or hemispheres; 10 studies (27.8%) presented country-based data (e.g., from the United States, Canada, China, and France). The remaining publications represented regional-level data with a relatively small scope, such as from Hong Kong SAR, People's Republic of China; Okinawa, (Japan); Madagascar; and French Guiana. A longitudinal study design combined with time-series analysis and statistical modeling was used in most of the included studies. The size of study population (if stated) ranged from 900 (21) to 5.4 billion (22). In 14 studies, authors used international databases such as FluNet ($n = 10$) and data from WHO–Global Influenza Surveillance and Response Network, whereas others used national influenza surveillance data. Data reported by sentinel general practitioners and hospital-based data were used in a few studies.

In terms of the outcome measurement, a large proportion ($n = 29$ of 36, 80.6%) of the included publications specified laboratory-confirmed cases to ascertain the outcome. Most of the included studies used time series, autoregressive integrated moving average, autoregressive models, mathematical modeling, geographic information systems, and Epi Map (US Centers for Disease Control and Prevention, Atlanta, GA) to establish or simulate spatiotemporal patterns of influenza epidemics. In many of the studies, multivariate modeling (e.g., linear regression, generalized linear models, logistic regression, other nonparametric approaches) and correlations (namely, Pearson and Spearman correlations) were performed to determine the contributing viral strains and effects of climatic factors on influenza seasonality.

The key findings of this review in relation to the study aims are organized into 3 major themes: 1) geographical and temporal variations in seasonal influenza; 2) influenza viral types in relation to spatiotemporal variations in influenza seasonal epidemics and cocirculation patterns of influenza viruses; and 3) the role of climatic factors in influencing spatiotemporal patterns of seasonal influenza.

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Theme 1: geographical and temporal variations in seasonal influenza

Seasonal influenza epidemics vary in their time of onset, duration, amplitude, and transmission patterns. These patterns were discussed in 23 of the 36 studies (63.9%) (10, 11, 14, 17–19, 22–38). The common observations and variations in these patterns were noticed at latitudinal levels according to temperate, tropical, and subtropical regions, as described in the next paragraph. The comparisons of the common patterns in different global regions are summarized in Web Table 1 (available at https://academic.oup.com/aje).

Geographic and temporal patterns in temperate regions. Influenza activity in temperate regions commonly peak during cold winter months across the globe (10, 11, 22, 24, 27). They begin in the northern hemisphere and spread southward, coinciding with the winter seasons of each hemisphere (19, 22, 27). In the northern hemisphere, influenza activity increases during January–March and in the southern hemisphere, the activity peaks in July–September. Influenza peak timing is associated with latitude in temperate regions of both hemispheres (10, 19). There is a lag of approximately 2.4 months for every 33° change in latitude from north to south (24). The average length of influenza season lasts approximately 4 months in temperate regions (10, 11, 19, 22, 36, 38). The amplitude of influenza transmission increases with increasing latitudes, particularly in the northern hemisphere, but such a variation is not observed in the southern hemisphere (10, 22, 24). Most temperate countries experience a single annual peak, usually during the coldest month of the winter (22). A horizontal pattern of influenza transmission occurs in some temperate countries in the northern hemisphere, such as Canada and in parts of Europe, with influenza cases emerging earlier in the western regions (18, 28).

Geographical and temporal patterns in tropical and subtropical regions. Many recent longitudinal studies (published in 2015 and 2016) were designed to address the knowledge gaps in understanding the patterns of influenza seasonality in tropical and subtropical regions with less well-defined but diverse seasonal patterns, complex climate conditions, and relative insufficient information (e.g., undersampled and unavailable data in many African, Central and South American, and Pacific Island countries) (10, 11, 20, 23, 29, 33, 39). The classification of focused regions differed in many studies. For example, some researchers specified the regions according to specific latitudinal gradients and classified the analyzed countries into tropical and subtropical regions (22, 33), whereas others combined tropical and subtropical regions in their analyses (10, 11, 19, 25).

The seasonal patterns of influenza are less prominent in tropical and subtropical regions. The onset of influenza season in these regions varies between the countries from April to June, with no distinct seasonal patterns. Some countries have biannual peaks and others have year-round influenza activity without noticeable peaks (10, 22, 29). The duration of influenza epidemics is longer in the tropics and subtropics compared with temperate regions, averaging 6 or 7 months (11). The amplitude of the seasonal influenza epidemics decreases toward the equator, especially in the tropics (10, 24, 29). Although the seasonal amplitude of influenza activity varied with the change in latitude in the northern hemisphere, such a relationship was not found in the southern hemisphere. This variation might be due to inadequate information available or because only a few

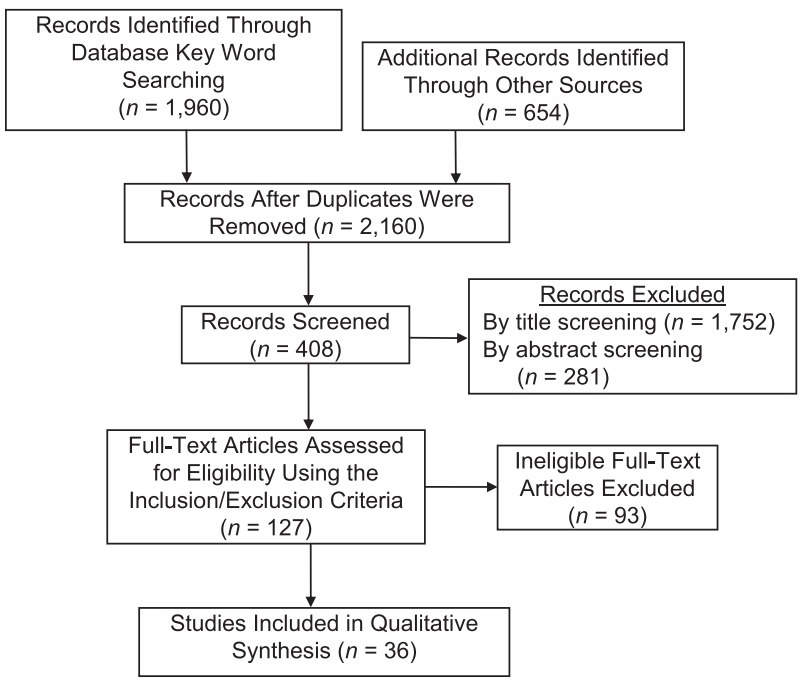


Figure 1. Flow diagram of the literature search and screening results.

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Table 1. Spatial and Temporal Patterns of Seasonal Influenza and Climatic Driving Factors Reported by the 36 Included Studies

First Author, Year (Reference No.)	Year of Study	Study Region/ Continent/ Country	Population	Study Type	Source of Data	Case Ascertainment	Statistical Analysis Methods	Spatial or Temporal Patterns	Associated Climatic Factors
Alonso, 2015 (23)	2002–2012	Antananarivo in Madagascar (in tropical region)	2 million in 2012	Longitudinal	National influenza center, FluNet, daily climate data from the International Research Institute for Climate and Society	Weekly reported and laboratory-confirmed influenza-positive cases	Time series analysis, nonparametric hierarchical clustering approach and Spearman correlations	Irregular timing of influenza activity and temporal patterns	Precipitation
Alonso, 2007 (24)	1979–2001	27 Brazilian states		Longitudinal	National Vital Statistics Agency for Brazilian Ministry of Health	Laboratory-confirmed influenza cases and death data	Detrended time-series analysis	Variations in the onset, duration, and amplitude of seasonal influenza in relation to latitude changes	Temperature and humidity
Azziz Baumgartner, 2012 (22)	1983–2008	85 countries globally	Cumulative population of 5.4 billion (83% of the world's population)	Longitudinal	WHO Global Influenza Surveillance Network, FluNet	Laboratory-confirmed influenza cases	Linear regression; χ^2 and Fisher exact tests	Different temporal patterns of influenza activity in temperate and tropical regions	Low temperature
Barreca, 2012 (40)	1973–2002	359 urban US counties		Longitudinal	National Center for Health Statistics	Influenza death data, climatic data	Nonlinear regression and correlations	Temporal variation and cross-state differences	Low temperature and low absolute humidity
Bedford, 2015 (25)	2000–2012	9,604 samples		Longitudinal	WHO–Global Influenza Surveillance and Response Network Data	Laboratory-confirmed cases	Epidemiologic modeling (model simulation)	Temporal circulatory patterns associated with influenza A subtypes H3N2 and H1N1, and influenza B	
Bloom-Feshbach, 2013 (10)	1990–2009	137 global locations		Literature review and longitudinal study	Existing literature and data from international surveillance websites	Laboratory-confirmed cases of influenza, published literature	Univariate and multivariate analyses using nonparametric Wilcoxon tests and linear regression	Variations in temporal patterns of influenza activity with the change in latitude and between temperate and tropical regions	

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Table 1. Continued

First Author, Year (Reference No.)	Year of Study	Study Region/ Continent/ Country	Population	Study Type	Source of Data	Case Ascertainment	Statistical Analysis Methods	Spatial or Temporal Patterns	Associated Climatic Factors
Caini, 2016 (11)	2000–2014	30 countries	571,907 cases	Retrospective study	Global Influenza B Study database	Weekly number of respiratory specimens tested for influenza viruses	Descriptive analyses of each study country/ region; Wilcoxon rank-sum test	Different temporal patterns in temperate and tropical regions; influenza A and B circulation overlaps in temperate regions	
Charland, 2009 (42)	2000–2005	35 cities in United States	63,334 cases	Longitudinal	Hospitalization records from 37 hospitals in the Pediatric Health Information System	A subset of ICD-9 codes used to identify influenza cases (identified from records of pediatric patients with ILI)	Bayesian hierarchical models	Timing and the peak of influenza activity associated with latitude	Solar radiation
Chong, 2015 (39)	2002–2009	Hong Kong		Longitudinal	Death data from Hong Kong Census and Statistics Department	Deaths due to influenza		Major peak of epidemic in winter	Cold temperature, rainfall
Deyle, 2016 (43)	1996–2014	26 countries		Longitudinal	FluNet, WHO global databases by country	Weekly laboratory-confirmed influenza cases	EDM; Pearson correlations; paired Wilcoxon tests; multivariate EDM approach; convergent cross-mapping	Spatial (latitudinal) variations and temporal patterns of influenza seasonality between temperate and tropical countries	Temperature and absolute humidity
Emukule, 2016 (26)	2007–2013	Kenya		Longitudinal	Data from 12 surveillance sites across Kenya	Laboratory-confirmed cases	Logistic regression and negative binomial regression methods	Temporal patterns in different regions; multiple epidemics each year	Low temperature and low specific humidity

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Table 1. Continued

First Author, Year (Reference No.)	Year of Study	Study Region/ Continent/ Country	Population	Study Type	Source of Data	Case Ascertainment	Statistical Analysis Methods	Spatial or Temporal Patterns	Associated Climatic Factors
Finkelman, 2007 (27)	1997–2005	19 temperate countries in northern and southern hemispheres		Longitudinal	FluNet, WHO global databases	Laboratory-confirmed influenza surveillance data by type and subtype	Correlations; independent *t* test (Student *t* test)	Varying temporal patterns in relation to different latitudes, hemispheres, and viral types	
He, 2013 (28)	1999–2012	Canada		Longitudinal	FluWatch surveillance program reports	Laboratory-confirmed influenza A cases	Generalized linear model; mathematical modeling	Spatial and temporal patterns	Low temperatures and low absolute humidity
He, 2015 (12)	2006–2015	138 countries		Longitudinal	Data from FluNet, WHO	Laboratory-confirmed cases (weekly data)	Time-series analysis; mathematical modeling	Global spatiotemporal patterns	
Hirve, 2016 (29)	2010–2014	70 countries	73% of the world's population	Longitudinal/ time-series data analysis	FluNet, WHO	Weekly proportional positive cases	Time-series analysis, binomial model	Spatial and temporal patterns in different continents and countries	
Iha, 2016 (30)	January 2007–March 2014	Okinawa, Japan	37,309 influenza A infections, 7,277 influenza B infections	Longitudinal	1. Data from 4 representative general hospitals; 2. Japanese national sentinel surveillance data	Weekly rapid antigen test results and nationwide surveillance reports	Spearman correlations	Temporal patterns with different viral strains	Low ambient temperature and relative humidity associated with influenza A epidemics; high relative humidity associated with influenza B epidemics
Jaakkola, 2014 (21)	July 2004–December 2005	Kajaani Garrison in northern Finland	892 military recruits	Case-crossover study	Follow-up data of the 892 Finn military recruits	Severe influenza cases (laboratory-confirmed)	One-way analysis of variance; conditional logistic regression models; discrete logistic model	Temporal patterns in subarctic climatic zone	Temperature and absolute humidity

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Table 1. Continued

First Author, Year (Reference No.)	Year of Study	Study Region/ Continent/ Country	Population	Study Type	Source of Data	Case Ascertainment	Statistical Analysis Methods	Spatial or Temporal Patterns	Associated Climatic Factors
Koul, 2014 (31)	2011–2012	New Delhi and Srinagar, India	Patients attending outpatient clinics in the study regions	Cohort study	Data of patients with influenza like illnesses	Laboratory-confirmed cases	Descriptive analyses of influenza data; correlations	Varying temporal patterns with latitude	Temperature, rainfall, and humidity
Mahamat, 2013 (45)	2006–2010	French Guiana	1,553 positive specimens	Longitudinal	Influenza surveillanc data collected by sentinel GP	Positive nasal swabs confirmed by laboratory testing	Univariate and multivariate analyses using ARIMA time-series analysis and dynamic regression	Seasonal circulations of pandemic influenza activity	Increased rainfall and specific humidity
N'Gattia, 2016 (46)	2007–2010	Abidjan, Cote d'Ivoire (African tropical region)		Retrospective time-series study	Data from influenza surveillance network	Laboratory-confirmed influenza cases	Box-Jenkin's method with ARIMA	Year round activity; no clear seasonality	Rainfall and relative humidity
Oliveira, 2016 (17)	2009–2013	Salvador, Brazil	2,651 patients ≥5 years old who attended emergency units in Salvador	Longitudinal	Data collected by Oswaldo Cruz Foundation as part of an enhanced surveillance study	Cases identified by blood tests and reviews of medical records	Negative binomial regression; χ^2 tests; Mann Whitney *U* test	Temporal patterns with annual peaks between August and September	
Paget, 2007 (18)	1999–2007	Members of European Influenza Surveillance Scheme for ≥5 years		Longitudinal	Data reported by sentinel GPs and national reference laboratories	Specimens of weekly sentinel reporting ILI and ARI cases were sent to reference laboratories for influenza infections	χ^2/Fisher exact tests and correlations.	Temporal and spatial patterns; length of epidemics correlated with different viral strains	

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Table 1. Continued

First Author, Year (Reference No.)	Year of Study	Study Region/ Continent/ Country	Population	Study Type	Source of Data	Case Ascertainment	Statistical Analysis Methods	Spatial or Temporal Patterns	Associated Climatic Factors
Roussel, 2016 (32)	2003–2013; excluding 2009–2010 due to pandemic	France		Longitudinal	Weekly reports from Regional Influenza Surveillance Group Network	Clinical reports by 25 sentinels and analysis of virologic samples	Principal component analysis; autoregressive linear model	Temporal and regional variations	Temperature, absolute humidity
Saha, 2016 (33)	2007–2013	9 countries in Southern and Southeast Asia	1,586,757 specimens	Longitudinal	Monthly data collected from 8 countries in the WHO Global Influenza Surveillance and Response System and data for Japan from FluNet	Patients with ILI or severe ARI were tested for influenza viruses (laboratory-confirmed cases).	Epi-Map used to plot seasonal patterns by country or region	Temporal patterns correlated with latitudinal locations and viral strains	
Shaman, 2010 (48)	1972–2002	United States		Longitudinal study and mathematical simulations		Influenza deaths 1972–2002	Epidemiologic model simulations based on previous laboratory findings	Spatial and temporal variations in the onset of epidemics and influenza seasonality in temperate regions	Absolute humidity
Soebiyanto, 2010 (49)	2005–2008	Hong Kong SAR, People's Republic of China; Maricopa County, Arizona		Longitudinal	Weekly influenza reports	Laboratory-confirmed cases	ARIMA and SARIMA (for including seasonality); correlations	Temporal patterns of influenza in tropical and subtropical regions	Rainfall, relative humidity, and temperature
Soebiyanto, 2014 (34)	2008–2013	Guatemala, El Salvador, and Panama		Longitudinal and statistical modeling	Influenza surveillance data (Panama); Laboratory data (Guatemala and El Salvador)	Laboratory-confirmed positive cases	Logistic regression	Temporal patterns of influenza activity	Specific humidity, temperature, and rainfall

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Table 1. Continued

First Author, Year (Reference No.)	Year of Study	Study Region/ Continent/ Country	Population	Study Type	Source of Data	Case Ascertainment	Statistical Analysis Methods	Spatial or Temporal Patterns	Associated Climatic Factors
Soebiyanto, 2015 (50)	2000–2011 for Spain; 2006–2011 for the other countries	Germany, Spain, Slovenia, and Israel		Longitudinal and statistical modeling	Influenza sentinel surveillance data	Weekly influenza cases, laboratory-confirmed samples	Generalized addictive model	Influenza seasonality in 3 temperate locations	Specific humidity, temperature, and solar radiation
Soebiyanto, 2015 (35)	2008–2013	Costa Rica, Honduras, and Nicaragua		Longitudinal and statistical modeling	National Influenza Center data (weekly samples tested positive for each influenza virus)	Laboratory-confirmed cases	Logistic regression	Changes in influenza activity in tropical locations associated with changes in climatic factors	Specific humidity, temperature
Suzuki, 2009 (41)	January 2001–March 2007	Okinawa, Japan		Longitudinal study	National Influenza Surveillance data	Laboratory-confirmed cases	Time-series analysis and phylogenetic tree analysis for viral types/subtypes	Regular winter peaks but summer peaks only in 4 of 6 seasons. The viral circulation in Okinawa does not always coincide with the patterns in mainland Japan	
Tamerius, 2013 (13)	1975–2008	40 countries globally		Longitudinal (analysis of developed global influenza database)	Data compiled on basis of published influenza surveillance studies as well as regional/ national surveillance schemes	Weekly or monthly laboratory-confirmed cases	Exploratory rank analyses and multivariate regression (logistic regression)	Influenza activity associated with latitude and climatic factors	Temperature and humidity

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Table 1. Continued

First Author, Year (Reference No.)	Year of Study	Study Region/ Continent/ Country	Population	Study Type	Source of Data	Case Ascertainment	Statistical Analysis Methods	Spatial or Temporal Patterns	Associated Climatic Factors
Tang, 2010 (36)	2000–2007	7 temperate and tropical locations in northern and southern hemispheres		Longitudinal	Weekly incidence data from WHO Collaborating Centers in 4 countries	Weekly incidence of influenza and climatic data	Time-series analysis using dynamic regression modeling approach and ARIMA	Temporal patterns in relation to temperate, subtropical and tropical regions, and viral types	Temperature and humidity
Thai, 2015 (20)	1993–2010	Vietnam	26,023,574 cases	Longitudinal study	Monthly ILI surveillance aggregated data from 52 provinces of Vietnam	Monthly influenza notifications	Time-series analysis and regression tree analysis	Variations in temporal pattern in northern and southern provinces of Vietnam	Absolute humidity
Weinberger, 2012 (37)	1915–2007	Iceland, North America, and Europe		Longitudinal study and time-series analysis	Disease surveillance records	Death and morbidity data from different countries	Serfling linear regression; Pearson correlations	Changes in influenza seasonality and timing of epidemics	
Members of Western Pacific Region Global Influenza Surveil-lance and Response System, 2012 (19)	2006–2010	14 countries with national influenza centers in Western Pacific region	196,720 specimens	Longitudinal	ILI and influenza virus data were obtained from the studied countries with national influenza centers and FluNet	Specimens processed for influenza virus testing to confirm cases	Descriptive analyses (tabulations and graphs)	Variations in timing of peak activity and temporal patterns between temperate and tropical regions, hemispheres, and different latitudinal locations.	
Yu, 2013 (38)	2005–2011	China		Longitudinal study	Weekly reports from sentinel hospitals in 30 provinces	Laboratory-confirmed influenza A and B infections	Joint seasonal regression; logistic regression	Geographical variations in influenza peak activity, duration, and periodic patterns	Low temperatures and increased rainfall

Abbreviations: ARI, acute respiratory infections; ARIMA, autoregressive integrative moving average; EDM, empirical dynamic modeling; GP, general practitioner; ICD-9, International Classification of Diseases, Ninth Revision; ILI, influenza-like illness; SARIMA, seasonal autoregressive integrative moving average; WHO, World Health Organization.

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studies addressed influenza patterns in the southern hemisphere (10, 29) (Web Table 1).

For larger countries, the temporal patterns of influenza epidemics vary between the different regions according to the latitudinal gradients (10, 11, 22–24, 29, 31, 33, 38). For example, the peak of influenza epidemics in China occur in January and February in the northern temperate provinces, whereas in the southern provinces (in tropical or subtropical regions), this peak occurs during spring and early summer in April–June. The provinces located between 27.4°N and 31.3°N have semiannual peaks, the first occurring during January and February and the second in June–August (38). Similar variations are also noticed in India, Brazil, Vietnam, and many African countries (20, 23, 24, 29, 31).

Countries situated in subtropical regions generally experience distinct, biannual influenza peaks. The first peak (usually a minor peak) generally occurs between December and March and the second peak (usually the major peak) emerges later in the year in June–October (in the subtropical north), coinciding with the monsoon seasons (11, 24, 29, 33). In tropical countries across the northern and southern hemispheres, influenza epidemics often exhibit year-round influenza activity or multiple peaks without distinct seasonal patterns (low seasonality). In addition, temporal characteristics such as the onset, duration, and peak timing of influenza epidemics are heterogeneous in tropical regions. The average duration of epidemics was slightly longer in the tropics (7–10 months) than in subtropical regions (5–6 months) (10, 11, 22, 29).

Theme 2: influenza viral types in relation to spatiotemporal variations in influenza seasonal epidemics and cocirculation patterns of influenza viruses

The viral strains responsible for seasonal influenza activity in humans include H3N2, H1N1, and influenza B viruses. The global circulatory patterns of influenza epidemics in relation to these viruses differ substantially in temperate, tropical, and subtropical regions. These variations were discussed in 12 of the 36 studies in terms of their time of onset, duration, cocirculation, and transmission patterns (11, 18, 19, 22, 25, 27, 28, 30, 33, 38, 40, 41). The common patterns are highlighted in the following paragraphs.

Influenza viral strains in relation to the patterns of seasonal influenza epidemics in temperate regions. H3N2 was the most dominant viral type in temperate regions of the northern and southern hemispheres, followed by influenza B and then H1N1 (27). In temperate regions, H3N2 epidemics have an earlier onset, approximately mid-January in the northern hemisphere and approximately June in the southern hemisphere. H1N1 epidemics in the northern hemisphere begin in late January, whereas in the southern hemisphere, the onset occurs in July or August (11, 25, 27). Influenza B activity is usually observed 4–5 weeks later, after the activity of either type of influenza A in the northern and southern hemispheres (22, 27, 28).

Although the duration of H1N1 epidemics in temperate regions is limited to the typical winter months, H3N2 and influenza B have substantial epidemic activity outside winter periods, extending to March in the northern hemisphere and to September in the southern hemisphere. There is a much temporal overlap between influenza A (either type) and influenza B epidemics in temperate regions (11, 27). For example, in a retrospective analysis of data from 30 countries over 14 years, researchers suggested the peak viral activities of influenza A and influenza B occur at the same time for approximately 40% of all observed epidemics in temperate regions of the northern hemisphere and for approximately 29% of those in the southern hemisphere (11). In another study, in which the patterns of influenza seasonality in 19 temperate countries from 1997 to 2005 were examined, researchers established a relationship between the timing of seasonal influenza epidemics and latitudinal gradients in temperate regions for influenza A. However, such a relationship was not observed for influenza B (27).

Influenza viral strains in relation to the patterns of seasonal influenza epidemics in tropical and subtropical regions. In tropical and subtropical regions, H1N1 and H3N2 epidemics are reported to have biannual activity for some regions and year-round activity for others. For example, in subtropical regions such as Hong Kong, there are 2 peaks of influenza epidemics: late February and early March and June or July (39). In another study, conducted on Okinawa, a subtropical island of Japan, similar patterns were identified: influenza A seasons often peaked in December–March, whereas influenza B epidemics followed influenza A seasons, peaking in March–July. Additional analysis of the temporal circulatory patterns of H1N1, H3N2, and Influenza B revealed that all 3 viral strains had increased activity during March–July in subtropical regions. This period represents the time between the influenza seasons in the northern and southern hemispheres (11, 25, 30).

For tropical countries like Singapore, influenza A epidemics occur sporadically throughout the year without any clear seasonal patterns (36). In the tropics, influenza B activity is reported year-round (with no prominent seasonality), particularly from January to August, whereas in subtropical regions, influenza B activity is frequently observed in March through to July (11, 23, 30). Specific seasonal patterns of influenza in relation to subtypes A and B were described in a study conducted by Saha et al. (33) in 8 Asian countries across a broad latitudinal range. They showed that influenza A and B epidemics coincided in regions north of 30°N during winter months. However, the majority of influenza A epidemics in regions 10°N–30°N occurred during monsoon months, whereas influenza B had a year-round pattern, with smaller peaks during the post monsoon months in areas south of 30°N (33). In tropical regions, the epidemics of influenza A and influenza B coincide less frequently than they do in temperate regions and usually peak at different times. Influenza B epidemics usually occur after influenza A epidemics in these regions. These features would probably extend the influenza seasons in the intertropical belt (11, 27, 30).

The average duration of the influenza epidemics is approximately 7 months for an influenza A season alone and approximately 8 months for the seasons in which

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influenza A and influenza B cocirculate (11). Bedford et al. (25) proposed that the long-lasting viral strains in tropical and subtropical regions may result in winter epidemics in temperate regions.

Subtype interactions between influenza A and influenza B viruses. Although H3N2 dominates in most of the seasonal epidemics in the northern and southern hemispheres, more than 1 subtype is usually reported during each season, suggesting that viral strains cocirculate in influenza epidemics (11, 27). As described previously, influenza B is usually reported to cocirculate with either H3N2 or H1N1. H3N2 and H1N1 provide cross-immunity for each other; therefore, these 2 viral subtypes rarely codominate in the same season (27). This pattern was also observed during the period after the 2009 influenza pandemic, in a study conducted by He et al. (12). The researchers noticed a period of reduced H1N1 activity after the pandemic in most European and East Asian countries (in 2009–2010 and a single wave in 2010–2011), whereas H3N2 was dominant during this "skip period" of H1N1 activity in 2011–2012. A resurgence of H1N1 was recorded in the 2012–2013 season. He et al. (12). also noticed that the onset of influenza B activity always lagged behind the H1N1 or H3N2 seasons irrespective of the pandemic. This finding was consistent with the results of most studies discussed in the previous section of this article. Saha et al. (33) investigated the patterns of influenza A and B epidemics before and after the 2009 pandemic (2007–2013) and supported the finding that H1N1 pdm (influenza A post-pandemic subtype) and H3N2 were the predominant subtypes contributing to the epidemics during 2009–2011 and in 2012 respectively. Nevertheless, Saha et al. (33) found that both subtypes were codominant in the studied Asian countries in 2013. This result contradicted the skip-year theory and existing evidence (12, 27). Bedford et al. (25) suggested that, unlike H1N1 and influenza B persisting across multiple seasons, H3N2 viruses seldom persisted locally over several epidemics but might be reseeded from East and Southeast (E-SE) Asia. Most importantly, H3N2 viruses were disseminated between global regions more frequently than were H1N1 and influenza B. Furthermore, global H3N2 epidemics were usually due to the introduction of new genetic variants in E-SE Asia, especially via overlapping epidemics of virus circulation (25). In a multicountry study in which the virological characteristics of influenza in the Western Pacific Region were investigated, WHO researchers, from 2006 to 2010, revealed that the dominant influenza strains in E-SE Asian countries were later detected in other countries. In this region, the majority of viruses identified between 2006 and 2009 were influenza A. However, an equal proportion of influenza A and B cases were reported in 2010. A dramatic increase in the number of influenza B specimens between 2009 and 2010 was noted. A detectable epidemic of influenza B (coexisting with H1N1 pandemic and H3N2, respectively, at early and later stages of the season) was observed across 4 latitudinal regions in the northern and southern hemispheres (19). The findings of these studies suggested that new genetic variants, possibly originating in E-SE Asian countries, particularly H3N2 and its cocirculation with influenza B, could potentially lead to large-scale epidemics.

Theme 3: the role of climatic factors in influencing spatiotemporal patterns of seasonal influenza

Temperature, humidity, and rainfall were repeatedly discussed as the major climatic and environmental factors that drive the major spatiotemporal patterns of seasonal influenza in different regions of the world, These factors were reported in 24 of the 36 studies in this review (20–22, 26–32, 34–36, 39, 40, 42–50). These climatic factors are also associated differently with the different viral strains.

The role of climatic factors in affecting the seasonal influenza patterns in temperate regions. According to Tamerius et al. (13), cold, dry climatic conditions drive the patterns of seasonal influenza in temperate regions. Low temperatures and absolute humidity (AH) are strongly associated with influenza activity in temperate regions (13, 21, 40, 48, 50). The decrease in AH increases the chances of viral survival and transmission, thus contributing to the increase in influenza activity (48). Jaakkola et al. (21) suggested the decrease in temperature together with low AH was more likely to contribute to the change in influenza activity rather than the actual low temperature. According to their study, a decrease in average temperature by 1°C increased the risk of influenza activity by 11%, and a decrease in AH of 0.5 g/m^3 increased the risk by 58% (21). Apart from temperature and AH, solar radiation was negatively associated with influenza seasonality in temperate regions (13, 39, 50).

Climatic factors contributing to the seasonal influenza patterns in tropical and subtropical regions. A higher level of influenza activity is reported to occur with an increase in rainfall in many tropical and subtropical countries (28, 43, 45, 46). Although low temperatures are strongly associated with winter onset of influenza epidemics in temperate regions, the role of temperature and humidity in tropical and subtropical regions is unclear. In some studies, it was suggested that increased rainfall and decreased specific humidity may be drivers for influenza seasons in the tropics and subtropical regions (13, 34, 43, 45). However, AH also had a U-shaped effect on influenza transmission in different regions (43). In the colder climates of temperate regions, lower AH promotes viral transmission, whereas in the warmer climates of tropical and subtropical regions, the increase in AH is associated with increased incidence of viral transmission (43). Rainfall and precipitation are positively associated with influenza activity in countries near the equator (23, 34, 35, 45, 46). In climatically diverse countries like Kenya, influenza activity increases with increased temperature and specific humidity; however, no important association was found between influenza activity and rainfall (26). The climatic drivers responsible for influenza A or influenza B epidemics are not well understood. Iha et al. (30) reported that low temperature and low relative humidity were related to influenza A activity, and high relative humidity was associated with influenza B epidemics in a subtropical setting (Okinawa) of Japan.

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DISCUSSION

The findings of this review have suggested the complex dynamics of the influenza seasonal patterns and their relationships with climatic factors. Seasonality of influenza is well defined in temperate regions but varies substantially in the intertropical belt. There were some common observations, such as winter epidemics in temperate regions, biannual or year-round activity in the tropics and subtropics, and north to south transmission of influenza, noted in many included studies. Overall, the general temporal patterns in different geographical locations identified in smaller individual studies were concordant with the results from multicountry studies in which databases at global level were used. However, the overall mechanism and factors driving these variations in the spatiotemporal patterns observed in different regions were still not well explained. In addition, heterogeneous temporal patterns of influenza seasonality existed within large countries and between countries in the same geographical regions, especially the countries situated within similar latitudinal ranges of tropical and subtropical regions such as southeast Asian, Central and South American, and African countries (11, 20, 22, 24, 26, 29, 33). Slightly different findings were reported in some global-scale longitudinal studies (using WHO FluNet surveillance data; comparisons of key findings are summarized in Web Table 2), particularly relating to peaks of influenza epidemics. For instance, Hirve et al. (29) revealed that unlike most Central and South American countries that have a single, distinct influenza peak, Belize, Colombia, Costa Rica, Ecuador, and Paraguay appeared to have 2 influenza peaks. However, Caini et al. (11) identified 2 annual influenza activity peaks in Nicaragua and Honduras. There were also inconsistent findings with regard to the common epidemic peaks in south and southeast Asia. Bloom-Feshbach et al. (10) stated that semiannual epidemics were observed predominantly in E-SE Asia, whereas Hirve et al. (29) found that most countries in the area experience a single distinct peak (with the exception of some countries situated in tropical or subtropical regions or near the equator). Interestingly, another study in which WHO influenza surveillance and FluNet data of 85 countries were used, researchers found that most Asian countries had year-round influenza activity (22). It also was reported in this study that countries in subtropical regions experienced a single influenza peak of epidemics coinciding with the coldest winter months each year, which was inconsistent with the common patterns observed in the present review. The inconsistent findings could be due to the different coverage years of WHO surveillance data, differences in defining the secondary peak of epidemic or irregular patterns, variations in analysis methods, and diversity in the quality and availability of surveillance data contributed by small but resource-restricted countries in those specific regions. Strengthening capacity of influenza surveillance in low-income countries is crucial to monitoring influenza transmission patterns and timing response to potentially large outbreaks. Researchers are urged to address the aforementioned variability in spatiotemporal patterns of influenza epidemics in those tropical and subtropical regions.

Although 2 distinct patterns in influenza activity were observed in this review according to geographical locations, a large proportion of the differences also depends on other factors, such as the dominant viral strain during a particular season, climatological conditions, the mode of viral transmission, and viral survival. The possible mechanisms are discussed in the following paragraphs in relation to different geographic regions.

The dry-cold climate driving influenza activity in temperate regions

Considering that influenza activity in temperate regions peaks during winter months in the northern and southern hemispheres, it is evident from this review that low temperatures and low AH are the major drivers of seasonal influenza activity in these regions. The "dry-cold mechanism" has been proposed by several researchers to explain the winter onset of influenza in temperate regions (13, 14, 21, 40, 47). According to the experimental studies conducted by Lowen et al. (44, 51), viral transmission of H3N2, H1N1, and influenza B is highly efficient at approximately 5°C and it is almost blocked at approximately 30°C. Lowen et al. also observed that the experimental animals shed viral particles more rapidly at low temperatures, particularly approximately 5°C (44, 51). Winters in temperate regions are characterized by average temperature between 0°C and 20°C, with the minimal temperature dropping to as low as −40°C in some parts of the regions. Thus, the suitable temperature range activating influenza viral transmission could partially explain the common winter epidemics in those regions.

Apart from the influence of temperature, the change in AH is considered another key contributor to the winter peak in influenza activity in temperate regions. Shaman et al. (47) found that 50% and 90% of variability in influenza viral survival and transmission, respectively, can be explained by the change in AH. They also suggested the reduction in AH in winter is consistent with a wintertime increase in influenza viral survival and transmission in temperate regions (47). Some live viral particles suspended in the air can be inhaled (i.e., aerosol transmission) by humans, and some can be transmitted through infected individuals (i.e., direct contact) or contaminated objects (i.e., indirect contact) (52–55). Weber et al. (56) explained the rate and mode of transmission would depend on the size of viral particles. Experimental studies conducted by the same researchers also revealed a decrease in relative humidity by 20% makes it more likely for small droplet particles to remain airborne for a significantly longer time—up to 24 hours—thus making the viruses easier to be spread to susceptible hosts through aerosol transmission (56). With supported evidence of these animal experiments, the dry-cold mechanism provides a reasoning framework to explain the interactions of particular climatic conditions and viral activity to enable large-scale transmission in cool seasons in temperate regions.

The other features of seasonal influenza in temperate regions identified in this review were periodicity and variations in annual occurrence of influenza epidemic relative to different viral strains. Temperate regions experience a single

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annual peak, usually during the same months every year, and the amplitude of influenza activity increases with latitude. The periodic pattern of influenza epidemic could be largely supported by the dry-cold theory (13). However, the existing literature remains inconclusive in establishing a comprehensive underlying mechanism to illustrate the geographic (i.e., latitudinal) differences of seasonal influenza in relation to different viral types.

The "humid-rainy mechanism" driving influenza seasonal patterns in tropics and subtropics

In tropical and subtropical regions, there is no clear seasonal pattern of influenza epidemics; however, most of these regions experience increased influenza activity in the monsoon season (31, 39, 45, 46). A large variation in influenza seasonal patterns is observed in these regions, with some countries having semiannual peaks in winter and spring, some having year-round influenza activity with peaks around the rainy seasons. There is a climatic diversity in these regions, with some countries being hot and dry, others being hot and humid. Although influenza epidemics in tropical regions have been a major focus of recent studies, existing evidence is limited to explain the heterogeneity in the influenza seasonal patterns in these regions and countries. One of the possible explanations regarding increased influenza activity associated with the monsoon seasons of different countries is the changes in the mode of viral transmission and viral survival during monsoons. According to results of studies of climatic parameters affecting influenza seasonality, ambient temperature and humidity may play an important role in viral transmission and viral survival, depending on the size of the viral particles (44, 56–58). Lowen et al. (51) demonstrated in their experiments that increasing ambient temperature in a laboratory setting increased the chance of contact transmission of influenza among guinea pigs but prevented airborne transmission. Contact transmission of influenza viruses is most efficient at 30°C and is not affected by the change in relative humidity (51). The results of another study indicated that wet conditions in the tropics would facilitate contact transmission by increasing the amount of viruses being deposited in fomites and by increasing viral survival in droplets (59). Considering that tropical countries in E-SE Asia are densely populated, contact transmission is likely to be the main mode of influenza transmission, particularly during rainy seasons, when people are likely to spend time indoors. However, this still does not explain the variations in the time of onset and amplitude of influenza activity according to the latitudes in tropical regions.

Furthermore, influenza viruses undergo 2 types of genetic change. A small, rapid genetic change called antigenic drift produces viral strains that are closely related to one another and are generally responsible for seasonal influenza. In contrast, a gradual major genetic change called antigenic shift produces a completely new strain of virus and is more likely to cause a global pandemic (60). Several studies on global phylogenetic analysis of influenza viral strains indicate that H3N2 viruses have a higher reproduction rate and evolve more rapidly compared with H1N1 and influenza B (25, 27, 61, 62). On the other hand, H1N1 is more likely to cause global pandemics than H3N2, particularly when it undergoes a major antigenic shift, as seen in the 1918 Spanish flu and the 2009 Swine flu pandemics (63). In this review, we also have highlighted the long duration of influenza B season in tropical and subtropical regions, especially when cocirculating with influenza A subtypes, and the increasing intensity of influenza B epidemics reported in southeast Asian countries. Most important, the dominant influenza strains in southeast Asian countries were later detected in other countries (19, 33). As mentioned previously, new genetic variants of influenza are likely to originate in E-SE Asia and potentially spread to other regions of the world (25). Furthermore, the atypical influenza seasonality (i.e., multiple peaks or year-round activity) and overlapping epidemics due to cocirculation of influenza viruses in this region may lead to further dissemination of influenza across regions or even continents. These findings warrant long-term monitoring of influenza activity in E-SE Asian countries, particularly during monsoon months in March–July to inform vaccination programs. In addition, the complex patterns of influenza seasonality in tropical and subtropical regions, the role of multistrain dynamics, and possible climatic drivers remain unclear (14, 27). Global efforts of influenza control and prevention and future studies can be devoted to a better understanding of global patterns of influenza epidemics in tropical and subtropical regions and the interaction of viral circulation and contributing climatic factors. It is also important to strengthen influenza surveillance systems in low- and middle-income countries, which usually are underrepresented in global studies, to produce quality adequate data for timely response to potential epidemics. It is also worth exploring the effects of climate change and interactions of meteorological factors on potential antigenic drift of influenza viral strains in different latitudinal regions. Other commonly identified contributors such as international travel, school and holiday periods, population immunity (e.g., vaccination), and susceptible subpopulations should be taken into consideration in studies.

This systematic review included a broad spectrum of longitudinal influenza studies covering billions of cases in more than 100 countries around the globe to provide a comprehensive insight on the patterns of seasonal influenza from existing evidence and possible mechanisms underpinning the complexity of the global influenza patterns. Despite the breadth and richness of population-based data included in this review, variations in influenza surveillance systems and case reporting could potentially limit the generalizability of the research findings. For example, many studies used data from global influenza surveillance networks such as FluNet and WHO databases, whereas others analyzed patient data from hospitalization records or reported by sentinel general practitioners. The use of country-aggregated data might have lost within-country variability in influenza spatiotemporal patterns, especially for large countries. Limitations of this study also include the differences in the definitions of influenza season and epidemic, peaks of epidemic, case reporting required in

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various surveillance systems, as well as between weekly or monthly reported cases. For example, weekly reported laboratory-confirmed influenza cases were analyzed in most studies, but in some studies, cases of laboratory-confirmed influenza and weekly reported influenza-like illness were combined, and influenza mortality cases were used in others. In this systematic review, we focused on describing the common spatial and temporal patterns of seasonal influenza and exploring possible influencing climatic factors and mechanisms proposed by empirical studies to explain the patterns. The review does not involve meta-analysis of collected data from the included studies to further establish statistical significance of the combined data. Influenza A and influenza B respond to changes in temperature, humidity, and rainfall differently; thus, a more detailed analysis of their circulation patterns in relation to these climatic factors could help us better understand the variations in influenza seasonality in temperate and tropical regions. Future studies may draw on the major findings of this systematic review to examine the quantitative associations of climatic and environmental factors with seasonal influenza driven by different viral types.

CONCLUSION

In conclusion, we have presented the common and different geographical and temporal patterns of seasonal influenza in relation to different viral strains in temperate, tropical, and subtropical regions through a systematic review and synthesis of findings from existing influenza studies. We also identified the climatic factors contributing to the spatiotemporal patterns of seasonal influenza in an attempt to identify the possible mechanisms to explain the variations in influenza seasonal patterns. The findings of this systematic review could inform the efforts of global surveillance of seasonal influenza and influenza prevention and control programs, such as vaccination recommendations, for different regions.

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